



(Operating manual)

# **ABRASION TEST MACHINE**

**Model: Ti/ABR.**



TESTMACH INTECH LLP  
VAPI-396195 (GUJARAT)  
CELL: +91 9924789891

E-mail: [prashant.patil@testmach.net](mailto:prashant.patil@testmach.net)

[support@testmach.net](mailto:support@testmach.net)

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## 1.MACHINE PHOTO:



## **2.PANEL DESCRIPTION :**

### **• MAIN POWER SUPPLY:**



*Back side view*

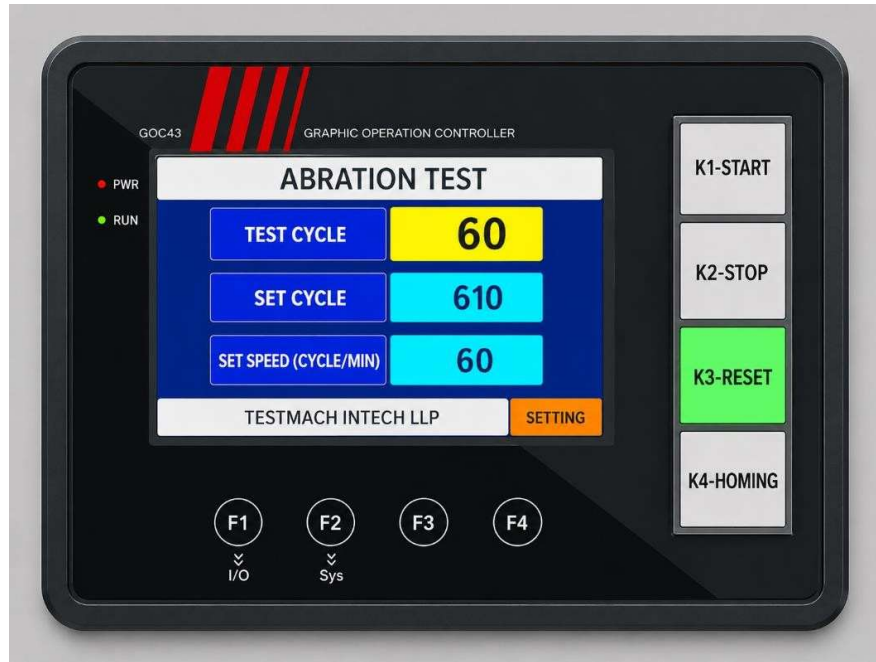
Power Description for Main Power Input Plug: The main power input plug should be connected as follows:

- L1 Single phases of the power supply.
- N is the neutral connection.
- E is the earth (ground) connection.
- The input voltage is 220-230V AC.

### **• PANEL FRONT & INSIDE PHOTO**



### 3.HMI TEST OPERATION:



| HMI SCREEN            |                                       | HMI Function Buttons |                                                                                         |
|-----------------------|---------------------------------------|----------------------|-----------------------------------------------------------------------------------------|
| TEST CYCLE            | Show Live Test Cycle                  | K1-START             | Test Start.                                                                             |
| SET CYCLE             | Set Test Cycles                       | K2-STOP              | Test Stop                                                                               |
| SET SPEED (Cycle/min) | Set Test Speed                        | K3-RESET             | Reset Test Cycle                                                                        |
| SETTING               | Go to Test Parameter settings Screen. | K4-HOMING            | 1. Machine start & go to home Position<br>2. Proxy Sense Indication (Green light Blink) |
|                       |                                       | F3                   | Test Speed increment.                                                                   |
|                       |                                       | F4                   | Test Speed Decrement.                                                                   |

**4.HMI TEST PARAMETER SETTING**



| HMI SCREEN            |                                     |
|-----------------------|-------------------------------------|
| SET CYCLE:            | Set Test Cycles                     |
| SET SPEED (Cycle/min) | Set Test Speed                      |
| CALIBRATION           | Go to Test Hardware Password Screen |
| HOME                  | Go to Home Screen                   |

## ➤ HMI TEST HARDWARE SETTING



### HMI SCREEN

**Enter Password 1001**

**Press OK, Go to Test Hardware Setting Screen**



## HMI SCREEN

|                                      |                                               |
|--------------------------------------|-----------------------------------------------|
| <b>SPEED CALIBRATION (Cycle/min)</b> | <b>Set Test Speed Calibration Default: 70</b> |
| <b>HOMING SPEED (Cycle/min)</b>      | <b>Set Test Homing Speed Default: 8</b>       |
| <b>SETTING</b>                       | <b>Go to Test Parameter Settings Screen</b>   |
| <b>HOME</b>                          | <b>Go to Home Screen</b>                      |

## **5.MACHINE PRE-INSPECTION (BEFORE TEST):**

- Check Power Supply & Emergency Pushbutton.
- Check Manually Machine (using 'Start' Button).
- Check Proxy for sensing (K4-Button Green LED)

## **6.MACHINE OPERATION (AUTO MODE):**

- Press the for. Button to position the machine at the home location.
- Properly install/place the sample into the machine.
- Set the preferred set cycle and set speed in the parameter setting
- Initiate the test by pressing the start button.
- The test will stop automatically after the specified number of cycles is completed.
- To pause the test midway, press the stop button.

## **7.ABRATION NEEDLE PIN :**

1. 150 gram (hand crome pin 1mm semi circle
2. Needle (25mm\*50mm\*10mm) @12mm felt
3. 150 Gram Pointer ( 1 deg.)

### **•Input Power Requirement & Wire Configuration**

#### **Input Power Requirement for ABRATION Machine:**

- **Input Supply:** 230V, Single-Phase, 50-60Hz, Neutral 3-Wire + Earthing
- **MCB Rating:** 10-16A, 3-Pole
- **Wire Specification:** 1.5-2.5 Sq.mm